🧬 Protein Structure Visualization Report

Hemoglobin (PDB ID: 1A3N)

📌 Title

Visualization and Structural Analysis of Hemoglobin using PyMOL

📖 Abstract

Hemoglobin is a vital oxygen-transport protein found in red blood cells. This project focuses on visualizing the three-dimensional structure of hemoglobin using computational tools. By analyzing its structure, important features such as alpha helices and functional regions can be identified, providing insights into its biological role.

🎯 Objective

- To visualize the 3D structure of hemoglobin
- To identify structural components such as helices and loops
- To highlight a specific region for structural analysis

🧰 Tools & Resources

- PyMOL
- RCSB Protein Data Bank
- Protein Data File: **1A3N.pdb**

🧪 Methodology

Step 1: Data Collection

The protein structure of hemoglobin was obtained from the Protein Data Bank using the PDB ID **1A3N**.

Step 2: Visualization

The downloaded .pdb file was opened in PyMOL for 3D visualization.

Step 3: Representation

- Cartoon representation was applied to observe overall structure
- Surface and stick models were used for detailed visualization

Step 4: Structural Coloring

- Alpha helices were colored red
- Beta sheets and loops were colored differently for clarity

Step 5: Region Highlighting

A selected region of the protein was highlighted using stick representation to demonstrate structural focus.

📊 Results

- The hemoglobin structure was successfully visualized in 3D
- The protein shows a complex folded structure dominated by alpha helices
- Multiple chains forming a tetrameric structure were observed
- A specific region was highlighted to demonstrate potential functional importance

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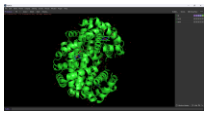
📸 Output Images



🧠 Discussion

Hemoglobin consists of four subunits forming a tetramer, which enables efficient oxygen binding and transport. The predominance of alpha helices contributes to its stability and function. Visualization of specific regions helps in understanding how structural variations may affect protein activity.

📚 Conclusion

This project demonstrates the importance of protein visualization in understanding molecular structure and function. Using PyMOL, complex protein structures like hemoglobin can be effectively analyzed and interpreted.

🔗 Data Source

Protein structure obtained from RCSB Protein Data Bank

👨‍💻 Author

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